

## KUOPIO RITONIEMI, Finland

WMO: 027320

Lat: 62.799N

Lon: 27.905E

Elev: 87

StdP: 100.28

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-27.5 | (c)<br>-24.1 | (d)<br>-30.8                  | (e)<br>0.2 | (f)<br>-27.6 | (g)<br>-26.9 | (h)<br>0.3 | (i)<br>-24.0 | (j)<br>11.0           | (k)<br>-1.5 | (l)<br>10.3 | (m)<br>-2.0 | (n)<br>1.2            | (o)<br>210 | (p)<br>0.668 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.5                    | 26.5            | 18.9 | 24.5 | 17.8 | 22.8 | 16.9 | 20.0                | 24.0 | 19.1 | 23.2 | 18.0 | 21.7 | 3.4                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.7                            | 13.7 | 22.0 | 17.5 | 12.7 | 21.1 | 16.3 | 11.7 | 20.1 | 57.8          | 23.5 | 54.6 | 23.3 | 51.1 | 21.6 | 23.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |    |    |
|-------------------|---------|---------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|----|----|
|                   |         |         | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     | DB | WB |
| (a) 10.3          | (b) 9.2 | (c) 8.2 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |
| (a) 10.3          | (b) 9.2 | (c) 8.2 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov  | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)  | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.3    | -9.8  | -7.3  | -3.9  | 2.6   | 9.9   | 14.5  | 17.7  | 16.5  | 11.4  | 4.4   | 0.0  | -5.2 |
|                                                                                  | DBStd       | 10.52  | 7.88  | 7.51  | 5.40  | 3.22  | 4.82  | 3.80  | 3.35  | 3.16  | 3.16  | 3.46  | 4.12 | 6.63 |
|                                                                                  | HDD10.0     | 2791   | 614   | 486   | 431   | 221   | 62    | 5     | 0     | 1     | 22    | 177   | 301  | 472  |
|                                                                                  | HDD18.3     | 5188   | 872   | 719   | 689   | 471   | 262   | 126   | 52    | 74    | 208   | 433   | 551  | 731  |
|                                                                                  | CDD10.0     | 708    | 0     | 0     | 0     | 0     | 59    | 140   | 239   | 202   | 65    | 2     | 0    | 0    |
|                                                                                  | CDD18.3     | 63     | 0     | 0     | 0     | 0     | 2     | 11    | 33    | 17    | 0     | 0     | 0    | 0    |
|                                                                                  | CDH23.3     | 302    | 0     | 0     | 0     | 0     | 14    | 49    | 167   | 72    | 0     | 0     | 0    | 0    |
|                                                                                  | CDH26.7     | 47     | 0     | 0     | 0     | 0     | 1     | 6     | 26    | 14    | 0     | 0     | 0    | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 3.9   | 4.0   | 4.3   | 3.7   | 3.6   | 4.1   | 3.5   | 4.0   | 4.3   | 4.8   | 4.6  | 4.4  |
| Precipitation                                                                    | PrecAvg     | 629    | 43    | 36    | 35    | 37    | 43    | 64    | 80    | 75    | 62    | 54    | 56   | 48   |
|                                                                                  | PrecMax     | 754    | 76    | 71    | 75    | 60    | 124   | 119   | 144   | 114   | 114   | 99    | 117  | 84   |
|                                                                                  | PrecMin     | 456    | 12    | 4     | 13    | 18    | 14    | 21    | 13    | 19    | 26    | 23    | 9    | 17   |
|                                                                                  | PrecStd     | 78     | 19    | 19    | 15    | 12    | 27    | 26    | 30    | 26    | 26    | 22    | 24   | 17   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 2.5   | 4.6   | 7.8   | 13.7  | 25.3  | 27.6  | 29.1  | 28.5  | 20.6  | 13.1  | 9.1  | 5.8  |
|                                                                                  |             | MCWB   | 1.6   | 2.6   | 3.6   | 7.8   | 15.1  | 18.8  | 19.6  | 20.3  | 15.6  | 10.7  | 7.7  | 4.9  |
|                                                                                  | 2%          | DB     | 1.7   | 3.1   | 5.3   | 11.4  | 22.8  | 24.7  | 27.1  | 25.1  | 18.1  | 11.6  | 7.4  | 3.9  |
|                                                                                  |             | MCWB   | 1.0   | 1.8   | 2.2   | 6.4   | 14.1  | 17.8  | 19.4  | 18.7  | 14.6  | 10.2  | 6.4  | 3.2  |
|                                                                                  | 5%          | DB     | 0.8   | 2.1   | 3.7   | 9.3   | 20.6  | 22.6  | 25.2  | 23.0  | 16.7  | 10.4  | 5.9  | 2.5  |
|                                                                                  |             | MCWB   | 0.2   | 1.3   | 1.3   | 5.3   | 13.8  | 16.5  | 18.5  | 17.3  | 14.0  | 9.3   | 5.5  | 1.6  |
|                                                                                  | 10%         | DB     | 0.1   | 1.1   | 2.5   | 7.6   | 17.7  | 20.7  | 23.3  | 21.3  | 15.6  | 9.2   | 4.7  | 1.4  |
|                                                                                  |             | MCWB   | -0.3  | 0.5   | 0.9   | 4.0   | 11.9  | 15.4  | 17.7  | 16.8  | 13.4  | 8.1   | 4.1  | 0.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 2.0   | 3.0   | 4.0   | 8.9   | 16.7  | 20.3  | 21.4  | 21.7  | 16.2  | 11.5  | 8.2  | 5.1  |
|                                                                                  |             | MCDB   | 2.2   | 4.1   | 6.4   | 11.2  | 22.8  | 24.6  | 26.1  | 27.6  | 18.4  | 12.5  | 9.2  | 6.0  |
|                                                                                  | 2%          | WB     | 1.2   | 2.0   | 2.7   | 7.0   | 15.2  | 19.0  | 20.3  | 19.3  | 15.5  | 10.5  | 6.5  | 3.2  |
|                                                                                  |             | MCDB   | 1.5   | 3.0   | 4.5   | 11.4  | 20.4  | 23.8  | 24.4  | 23.2  | 17.2  | 11.3  | 7.3  | 3.8  |
|                                                                                  | 5%          | WB     | 0.5   | 1.3   | 1.9   | 5.6   | 14.1  | 17.2  | 19.5  | 18.2  | 14.6  | 9.4   | 5.4  | 1.7  |
|                                                                                  |             | MCDB   | 0.9   | 1.9   | 3.2   | 8.4   | 19.6  | 21.5  | 23.5  | 21.8  | 16.3  | 10.3  | 5.9  | 2.3  |
|                                                                                  | 10%         | WB     | -0.2  | 0.6   | 1.2   | 4.5   | 12.5  | 15.7  | 18.7  | 17.3  | 13.7  | 8.1   | 4.2  | 0.9  |
|                                                                                  |             | MCDB   | 0.3   | 1.1   | 2.3   | 6.9   | 17.5  | 19.9  | 22.6  | 20.6  | 15.4  | 9.1   | 4.6  | 1.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.8   | 6.0   | 7.8   | 7.4   | 8.0   | 6.7   | 6.5   | 5.8   | 4.5   | 3.4   | 2.8  | 4.1  |
|                                                                                  |             | MCDBR  | 3.5   | 4.7   | 7.4   | 10.5  | 11.7  | 9.3   | 9.4   | 8.3   | 6.1   | 4.3   | 3.8  | 3.5  |
|                                                                                  |             | MCWBR  | 3.4   | 4.1   | 5.2   | 6.5   | 5.3   | 4.0   | 3.6   | 3.3   | 3.6   | 3.3   | 3.7  | 3.3  |
|                                                                                  | 5% WB       | MCDBR  | 3.3   | 4.3   | 6.0   | 8.9   | 11.4  | 8.6   | 8.3   | 7.2   | 5.6   | 4.3   | 3.8  | 3.4  |
|                                                                                  |             | MCWBR  | 3.3   | 3.9   | 4.6   | 5.8   | 5.4   | 4.0   | 3.9   | 3.4   | 3.6   | 3.7   | 3.7  | 3.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.231  | 0.287 | 0.313 | 0.347 | 0.345 | 0.343 | 0.367 | 0.351 | 0.328 | 0.313 | 0.249 | N/A  |      |
|                                                                                  | taud        | 2.009  | 2.115 | 2.186 | 2.347 | 2.468 | 2.499 | 2.447 | 2.478 | 2.531 | 2.400 | 2.100 | N/A  |      |
|                                                                                  | Ebn at Noon | 402    | 640   | 779   | 837   | 874   | 881   | 845   | 829   | 782   | 631   | 377   | N/A  |      |
|                                                                                  | Edh at Noon | 33     | 67    | 93    | 99    | 95    | 94    | 97    | 86    | 68    | 52    | 30    | N/A  |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.19   | 0.76  | 2.29  | 3.66  | 4.80  | 5.28  | 5.02  | 3.84  | 2.15  | 0.83  | 0.21  | 0.09 |      |
|                                                                                  | RadStd      | 0.03   | 0.18  | 0.28  | 0.43  | 0.53  | 0.32  | 0.50  | 0.44  | 0.24  | 0.14  | 0.04  | 0.01 |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (38 neighbors) | +0.55 | N/A     | N/A    | N/A     | N/A   | N/A   | -188        | N/A     | N/A     | N/A     |

Nomenclature: See separate page